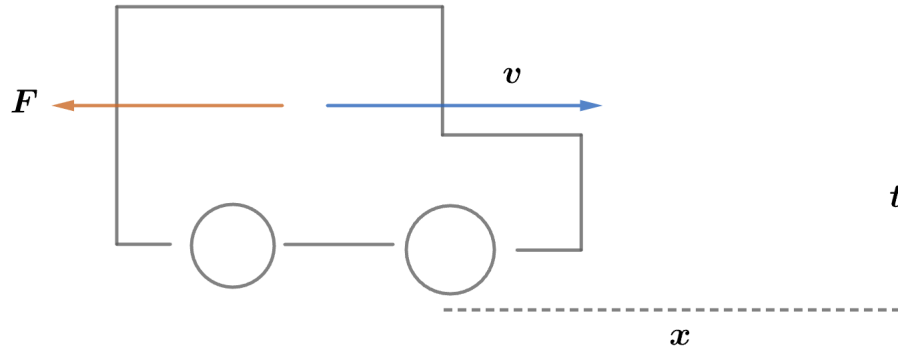


2013 F=ma Exam: Problem 9

Kevin S. Huang



From the previous problem, we found the initial kinetic energy is

$$K_i = \frac{1}{2}mv^2 = Fx$$

Solving for the mass,

$$m = \frac{2Fx}{v^2}$$

Then the initial momentum is

$$p_i = mv = \frac{2Fx}{v}$$

so the answer is E.